

Fractures, dislocations and sprains: Texts

Text A

Fractures (buckle or break in the bone) often occur following direct or indirect injury, e.g. twisting, violence to bones. Clinically, fractures are either:

- closed, where the skin is intact, or
- compound, where there is a break in the overlying skin

Dislocation is where a bone is completely displaced from the joint. It often results from injuries away from the affected joint, e.g. elbow dislocation after falling on an outstretched hand.

Sprain is a partial disruption of a ligament or capsule of a joint.

Text B

Simple Fracture of Limbs

Immediate management:

- Halt any external haemorrhage by pressure bandage or direct pressure
- Immobilise the affected area
- Provide pain relief

Clinical assessment:

- Obtain complete patient history, including circumstances and method of injury
 - medication history – enquire about anticoagulant use, e.g. warfarin
- Perform standard clinical observations. Examine and record:
 - colour, warmth, movement, and sensation in hands and feet of injured limb(s)
- Perform physical examination
 - Examine:
 - all places where it is painful
 - any wounds or swelling
 - colour of the whole limb (especially paleness or blue colour)
 - the skin over the fracture
 - range of movement
 - joint function above and below the injury site
 - Check whether:
 - the limb is out of shape – compare one side with the other
 - the limb is warm
 - the limb (if swollen) is throbbing or getting bigger
 - peripheral pulses are palpable

Management:

- Splint the site of the fracture/dislocation using a plaster backslab to reduce pain
- Elevate the limb – a sling for arm injuries, a pillow for leg injuries
- If in doubt over an injury, treat as a fracture
- Administer analgesia to patients in severe pain. If not allergic, give morphine (preferable); if allergic to morphine, use fentanyl
- Consider compartment syndrome where pain is severe and unrelieved by splinting and elevation or two doses of analgesia
- X-ray if available

Text C

Drug Therapy Protocol:

Authorised Indigenous Health Worker (IHW) must consult Medical Officer (MO) or Nurse Practitioner (NP). Scheduled Medicines Rural & Isolated Practice Registered Nurse may proceed.

Drug	Form	Strength	Route of administration	Recommended dosage	Duration
Morphine	Ampoule	10 mg/mL	IM/SC	Adult only: 0.1-0.2 mg/kg to a max. of 10 mg	Stat
			IV (IHW may not administer IV)	Adult only: Initial dose of 2 mg then 0.5-1 mg increments slowly, repeated every 3-5 minutes if required to a max. of 10 mg	Further doses on MO/NP order

Use the lower end of dose range in patients ≥ 70 years.

Provide Consumer Medicine Information: advise can cause nausea and vomiting, drowsiness.

Respiratory depression is rare – if it should occur, give naloxone.

Text D

Technique for plaster backslab for arm fractures – use same principle for leg fractures

1. Measure a length of non-compression cotton stockinette from half way up the middle finger to just below the elbow. Width should be 2–3 cm more than the width of the distal forearm.
2. Wrap cotton padding over top for the full length of the stockinette – 2 layers, 50% overlap.
3. Measure a length of plaster of Paris 1 cm shorter than the padding/stockinette at each end. Fold the roll in about ten layers to the same length.
4. Immerse the layered plaster in a bowl of room temperature water, holding on to each end. Gently squeeze out the excess water.
5. Ensure any jewellery is removed from the injured limb.
6. Lightly mould the slab to the contours of the arm and hand in a neutral position.
7. Do not apply pressure over bony prominences. Extra padding can be placed over bony prominences if applicable.
8. Wrap crepe bandage firmly around plaster backslab.

END OF PART A

THIS TEXT BOOKLET WILL BE COLLECTED

Part A

TIME: 15 minutes

- Look at the four texts, **A-D**, in the separate **Text Booklet**.
 - For each question, **1-20**, look through the texts, **A-D**, to find the relevant information.
 - Write your answers in the spaces provided in this **Question Paper**.
 - Answer all the questions within the 15-minute time limit.
 - Your answers should **only** be taken from texts **A-D** and must be correctly spelt.
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Fractures, dislocations and sprains: Questions

Questions 1-7

For each question, **1-7**, decide which text (**A, B, C** or **D**) the information comes from. Write the letter **A, B, C** or **D** in the space provided. You may use any letter more than once.

In which text can you find information about

- | | | |
|---|--|-------|
| 1 | procedures for delivering pain relief? | _____ |
| 2 | the procedure to follow when splinting a fractured limb? | _____ |
| 3 | what to record when assessing a patient? | _____ |
| 4 | the terms used to describe different types of fractures? | _____ |
| 5 | the practitioners who administer analgesia? | _____ |
| 6 | what to look for when checking an injury? | _____ |
| 7 | how fractures can be caused? | _____ |

Questions 8-14

Answer each of the questions, **8-14**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both. You should **not** write full sentences.

- | | | |
|----|--|-------|
| 8 | What should be used to elevate a patient's fractured leg? | _____ |
| 9 | What is the maximum dose of morphine per kilo of a patient's weight that can be given using the intra-muscular (IM) route? | _____ |
| 10 | Which parts of a limb may need extra padding? | _____ |

- 11 What should be used to treat a patient who suffers respiratory depression?
- _____
- 12 What should be used to cover a freshly applied plaster backslab?
- _____
- 13 What analgesic should be given to a patient who is allergic to morphine?
- _____
- 14 What condition might a patient have if severe pain persists after splinting, elevation and repeated analgesia?
- _____

Questions 15-20

Complete each of the sentences, **15-20**, with a word or short phrase from one of the texts. Each answer may include words, numbers or both.

- 15 Falling on an outstretched hand is a typical cause of a _____ of the elbow.
- 16 Upper limb fractures should be elevated by means of a _____ .
- 17 Make sure the patient isn't wearing any _____ on the part of the body where the plaster backslab is going to be placed.
- 18 Check to see whether swollen limbs are _____ or increasing in size.
- 19 In a plaster backslab, there is a layer of _____ closest to the skin.
- 20 Patients aged _____ and over shouldn't be given the higher dosages of pain relief.